$$\begin{aligned}
\rho_{\mathcal{S}}^{(01)}(t) &= \rho_{\mathcal{S}}^{(01)}(0)\,\mathrm{Tr}_{\mathcal{E}}\left\{\exp\left[\sum_i\left(\lambda_i(t)\hat{a}_i^\dagger - \lambda_i^*(t)\hat{a}_i\right)\right]\bigotimes_i \hat{\rho}_{\mathcal{E}_i}\right\} \\
&= \rho_{\mathcal{S}}^{(01)}(0)\prod_i \mathrm{Tr}_{\mathcal{E}_i}\left\{\hat{D}\left(\lambda_i(t)\right)\hat{\rho}_{\mathcal{E}_i}\right\} \\
&= \rho_{\mathcal{S}}^{(01)}(0)\prod_i \left\langle \hat{D}\left(\lambda_i(t)\right)\right\rangle_{\hat{\rho}_{\mathcal{E}_i}} = \left[\rho_{\mathcal{S}}^{(10)}(t)\right]^* . \qquad (5.117)
\end{aligned}$$